

Xiang He

Ph.D. Candidate in Geosciences, The University of Texas at Austin

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Research Profile

Computational geodynamicist studying how mantle flow, subduction, lithospheric deformation, and mantle rheology shape observable surface motions. I build physics-based numerical models and compare model predictions with geological and geophysical constraints, with current emphasis on Western U.S. vertical motion and Cascadia.

Education

The University of Texas at Austin

Ph.D. in Geosciences / Geophysics

Aug. 2024–Present

Austin, TX, USA

- Advisor: Prof. Thorsten W. Becker
- Research focus: computational geodynamics, mantle flow, subduction, dynamic topography, vertical crustal motion

University of Science and Technology of China

B.S. in Geophysics

Sep. 2020–Jun. 2024

Hefei, Anhui, China

- Guo Moruo Scholarship; China National Scholarship

Research Experience

The University of Texas at Austin

Graduate Researcher, Institute for Geophysics / Department of Earth and Planetary Sciences

Aug. 2024–Present

Austin, TX

- Develop global and regional geodynamic models to investigate mantle-flow contributions to long-wavelength surface deformation and vertical motion.
- Current project: construct Cascadia subduction and Western U.S. geodynamic models and evaluate predicted vertical motion against geophysical observations.
- Design numerical experiments to test sensitivity to slab geometry, mantle rheology, density structure, and boundary conditions.
- Run and analyze large-scale simulations using ASPECT/CitcomS and high-performance computing workflows; process model output with Python, GMT, and related tools.

University of Illinois Urbana-Champaign

Visiting Researcher, Department of Earth Science and Environmental Change

May 2023–Oct. 2023

Urbana, IL

- Conducted 3-D regional geodynamic modeling of intraplate deformation in East Asia under the supervision of Prof. Lijun Liu.

University of Science and Technology of China

Undergraduate Researcher, School of Earth and Space Sciences

Jan. 2023–Aug. 2023

Hefei, China

- Modeled planetary crust formation and thermal-mechanical evolution under the supervision of Prof. Wei Leng.

University of Science and Technology of China

Undergraduate Researcher, School of Earth and Space Sciences

Apr. 2022–Jun. 2024

Hefei, China

- Investigated plume–subduction interactions and the influence of African mantle dynamics on Tethyan evolution under the supervision of Prof. Jinshui Huang.

Publications

[1] Xiang He et al. "REPLACE WITH FULL TITLE". In: *Science* (2026). Accepted.

Selected Research Themes

- **Western U.S. vertical motion and Cascadia:** subduction-system dynamics, mantle flow, dynamic topography, and comparison with surface observations.
- **Mantle rheology and plate motion:** sensitivity of plate-scale deformation and surface observables to mantle viscosity structure and density heterogeneity.
- **Subduction and plume dynamics:** numerical models of single/double subduction, plume–plate interactions, and long-term tectonic evolution.

Technical Skills

Programming: Python, C, Fortran, MATLAB, shell scripting

Geodynamic / Geophysical Software: ASPECT, CitcomS, GPlates, GMT / PyGMT, ParaView

Scientific Computing: HPC workflows, MPI-based simulations, Linux, Git, numerical modeling, inverse problems, data–model comparison

Methods: Finite-element geodynamics, sensitivity analysis, geophysical data processing, visualization

Awards & Honors

Graduate Fellowship, The University of Texas at Austin	2024–2025
Guo Moruo Scholarship, USTC	2024
China National Scholarship (top 0.2% in China)	2023
Yang Yongman Scholarship, USTC	2023
USTC Outstanding Student Scholarship, Gold Medal (top 3%)	2022
USTC Outstanding Student Scholarship, Silver Medal (top 10%)	2021
Zhao Jiuzhang Earth Science Honored Class Scholarship	2021, 2022

Teaching

University of Science and Technology of China

Teaching Assistant, Electromagnetism A (PHYS1004A.04.2023SP)

Instructor: Prof. Gan Qin

Mar. 2023–Jun. 2023

Hefei, China